

TM Forum Component

Product Configurator

TMFC027

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Notice

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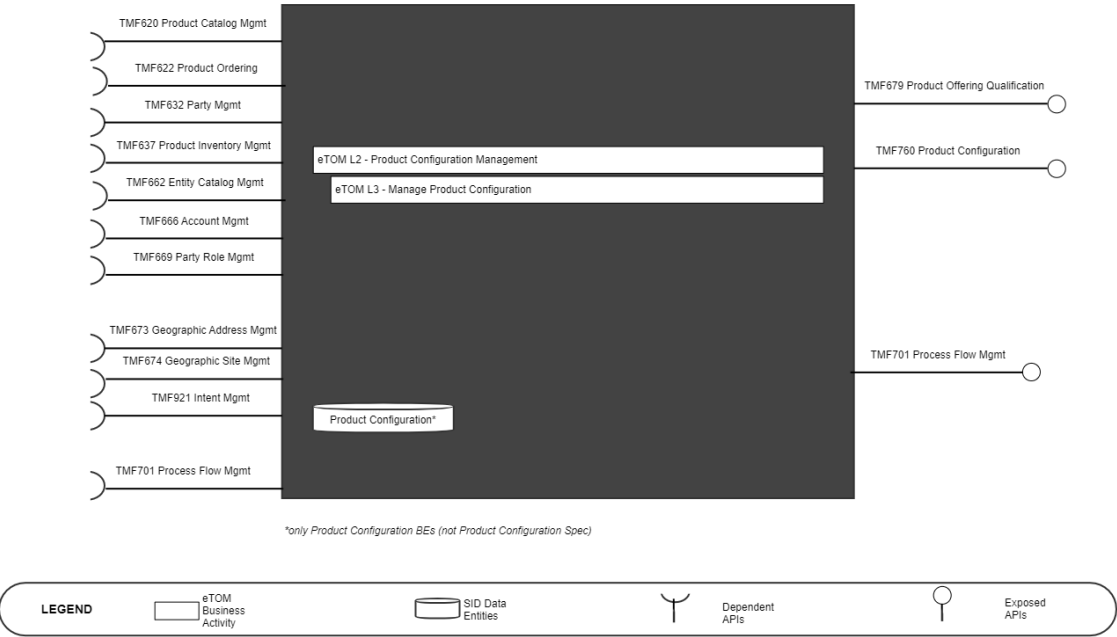
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1. Overview

Component Name	ID	Description	ODA Function Block
Product Configurator	TMFC027	The Product Configurator aims to provide sales representatives and customers with fast and error-free product qualification and product configuration capabilities across all channels. It uses mostly the Product Catalog and the Product Inventory information, and is able to execute and check all types of Product Catalog policy rules (e.g. packaging rules, commercial pre-requisite rules, pricing rules, consistency rules between characteristic values). It can also use Party related information, such as the age of the customer, or specific contextual information, such as the channel. It can be triggered in contexts such as product order capture, shopping cart or quote management. It supports the Product Order Capture and Validation component (TMFC002) to: • Establish commercial offering eligibility, and to provide alternative product offerings in case of ineligible product offering selection • Leads the configuration of selected eligible product offerings according to the context. Product configuration includes computation of characteristics values, allowed values, bundled and related product offering, pricing and discounts	Core Commerce Management

TMFC027: Product Configurator



2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM business activities

eTOM business activities this ODA Component is responsible for:

Identifier	Level	Business Activity Name	Description
1.2.5	L2	Product Configuration Management	Configuration Management configures or creates a new version of a configuration for an entity, such as a product, service or resource , as defined by a configuration specification. This process also modifies a configuration and values for configuration parameters, and removes a configuration.
1.2.5.2	L3	Manage Product Configuration	Manage Product Configuration business activity is in charge of creating, maintaining, controlling, changing and reporting Product Configuration according to Product Configuration Plans. Manage Product Configuration will establish and maintaining consistency of a product's performance, functional, and physical attributes within the limits defined by product requirements, product design, and operational information throughout the Products Lifecycle.

2.2. SID ABEs

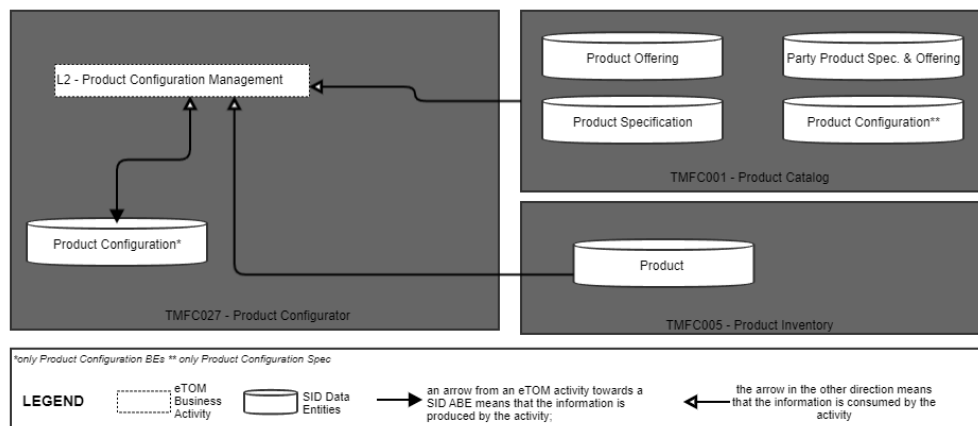
SID ABEs this ODA Component is responsible for:

SID ABE Level 1	SID ABE L1 Definition	SID ABE Level 2 (or set of BEs)*	SID ABE L2 Definition
Product Configuration	A Product Configuration (also referred to as a Product Profile) defines how a Product operates or functions. A Product Configuration may contain one or more parts, and each part may contain zero or more fields. Each field may have attributes that are statically or dynamically defined. Some of these fields have fixed values, while others provide values from which a choice or choices can be made (e.g. using the EntitySpec/Entity and/or CharacteristicSpec/CharacteristicValue patterns) ¹	ProductConfiguration	A representation of how a Product operates or functions in terms of characteristics and related Product(s).

¹ As in GB922 Product v23.0 document the definition of the ABE Product Configuration is a copy of the Configuration ABE pattern defined in GB922 Common v23.0, it is here adapted to the product level only.

*: if SID ABE Level 2 is not specified this means that all the L2 business entities must be implemented, else the L2 SID ABE Level is specified.

2.3. eTOM L2 - SID ABEs links



2.4. Functional Framework Functions

Function ID	Functional Framework Function	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
55	Price & Discount Calculation	Price and Discount Calculation applies pricing and discounting rules and algorithms in the context of the assembled information concerning Products (i.e. instances of Product).	Rating and Follow up	Tariff Calculation and Rating
182	Inter Product Dependency Identification	Inter-Product Dependency Identification identifies product dependencies, binds new order to purchased product or point to the dependent product required	Product Configuration & Activation	Offer and Product Configuration

Function ID	Functional Framework Function	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
205	Customer Order Eligibility Validation	Customer Order Eligibility Validation function validates that the Offer & products specified on the Customer Order, are eligible from a commercial and functional point of view. It includes: • Commercial Eligibility with commercial compatibility with the already customer installed Offers • Functional Eligibility with the customer's already installed Products (corresponding to ProductSpecification).	Customer Order Management	Customer Order Eligibility Validation
207	Offer and Product Configuration	The Offer and Product Configuration function enables the configuration of the commercial offer chosen by the customer. The configuration recovers the choice of an option, the choice of the characteristics values for the Product Specification including installation preferences... It can be based on product configurator using a rule engine. It can be used at the same time by Selling, Order Establishment or Develop Sales Proposal Business activities.	Product Configuration & Activation	Offer and Product Configuration
262	Product Availability Checking ¹	Product Availability Checking function provide an internet technology driven interface for the customer to undertake a product availability check. E.g., that the product offering is active for	Customer Order Management	Customer Order Eligibility Validation

Function ID	Functional Framework Function	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
		sales, the equipment(s) specified in the customer order are on stock.		
274	Quote Price Support Access	Quote Price Support Access provides self empowered fulfillment function to provide an internet technology driven interface for the customer to get a Quotation price.	Customer Order Management Fulfillment Integration Management	Customer Order Quotation Customer Fulfillment Access Management
278	Customer SLA Preferences Capturing	Customer SLA Preferences Capturing captures the customer's SLA preferences e.g., as part of the fulfillment.	Product Configuration & Activation	Offer and Product Configuration
300	Discount Calculation	Discounts Calculation determines charge discounts based on pricing plan; including discounts on recurring, one time, and usage charges. Discounts may be applied at different levels such as cross product, cross location, or cross customer (all customers that are part of a given group plan – some affiliation). The discounts can be apportioned across multiple events.	Rating and Follow up	Tariff Calculation and Rating
320	Customer Product Proposal Creation	Customer Product Proposal Creation proposes according to what the customer presently has as part of what can be further provided to the customer including bundling, product proposals, etc.	Product Configuration & Activation	Offer and Product Configuration
379	Product Customization Offering Management	Product Customization Offering Management provides the necessary functionality to manage the customer personalized proposals,	Sales Management	Opportunity Management

Function ID	Functional Framework Function	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
		taking into account the customer location, needs, current products, as well as the service provider's products, sales emphasis and targets, etc.		
727	Product Offer to Customer Verification ²	Product Offer to Customer Verification enables and verifies the configuration of the commercial offer chosen by the customer. The configuration consist of technical, functional, and commercial prerequisites and preferences.	Customer Order Management	Customer Order Eligibility Validation
928	Solution Design Creation	Solution Design Creation function combine/configure the emerged solution based on the existing solution, planned changes and newly designed features (e.g., for site connectivity services) Selecting relevant Products and Services from Catalog – Browse and select entries from a catalog that might be relevant to the set of captured requirements into the design.	Sales Management	Opportunity Management
930	Automatic Solution Validation	Obtains configuration constraints from the catalog and validates the correctness of the design against them.	Sales Management	Opportunity Management
931	Solution Pricing	The Solution Pricing function is concerned with assuring that the designs are priced consistent with pricing used for billing. The common product catalog provides an initial price	Sales Management	Opportunity Management

Function ID	Functional Framework Function	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
		base for the components that are in the solution vs. the existing configuration at the customer location.		
932	Calculation Rules Retrieval	Calculation Rules Retrieval function gives support for tariffication rules including discounting rules. It supports that discounting rules and guidelines are provided as to standard levels of discounts/promotions that can be provided to the customer. Special discount arrangements can be obtained by following an escalation process. There is workflow functionality to help manage discount escalation.	Sales Management	Opportunity Management
933	Price/Cost Optimization Price Optimization	Price Optimization enables sales to effectively evaluate the customer, generate recommendations for price decreases and increases, and set negotiation guidelines based on our cost. This includes the application of non-standard pricing.	Sales Management	Opportunity Management

¹ TMFC027 Product Configurator covers the part related to Product Catalog rules checking. Stock control part is done by TMFC002 Product Order Capture & Validation

² TMFC027 Product Configurator covers the part related to Product Catalog rules checking at commercial and functional eligibility levels. Technical eligibility controls are triggered by TMFC002 Product Order Capture & Validation

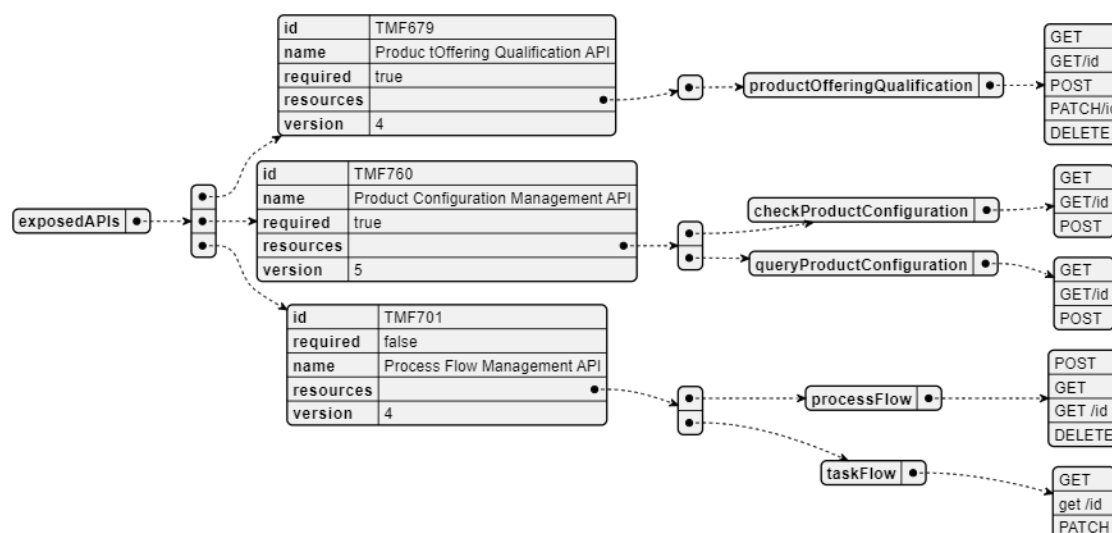
3. TMF OPEN APIs & Events

The following part covers the APIs and Events; This part is split in 3:

- List of **Exposed APIs** - This is the list of APIs available from this component.
- List of **Dependent APIs** - In order to satisfy the provided API, the component could require the usage of this set of required APIs.
- List of **Events (generated & consumed)** - The events which the component may generate is listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

3.1. Exposed APIs

Following diagram illustrates API/Resource/Operation:

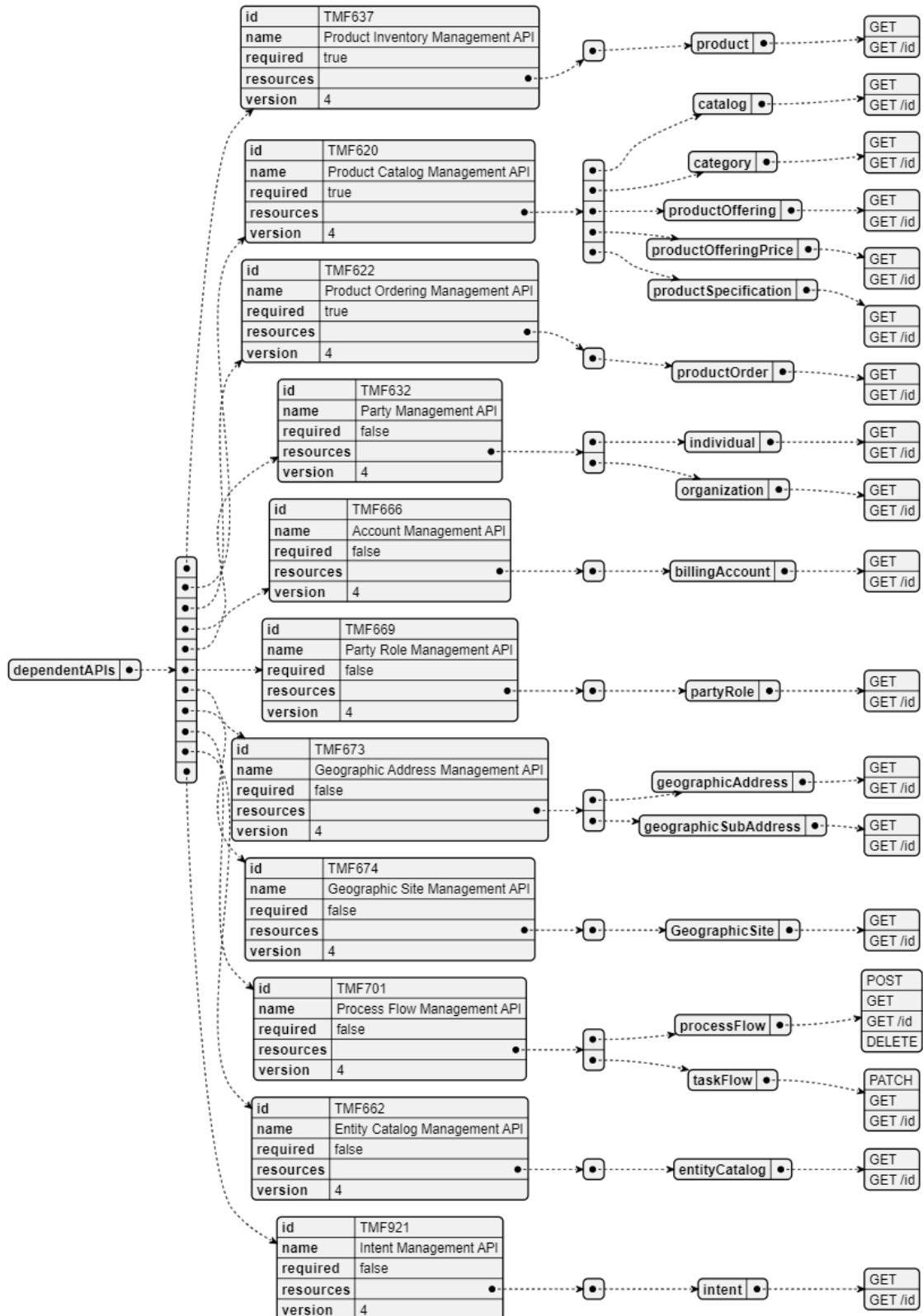


API ID	API Name	API Version	Mandatory / Optional	Operations
TMF679	Product Offering Qualification	V4	Mandatory	productOfferingQualification: • GET • GET /id • POST • PATCH • DELETE

API ID	API Name	API Version	Mandatory / Optional	Operations
TMF760	Product Configuration	V5	Mandatory	checkProductQualification: - GET - GET /id - POST queryProductQualification: - GET - GET /id - POST
TMF688 https://raw.githubusercontent.com/tmforum-apis/TMF688-Event/master/TMF688-Event-v4.0.0-swagger.json	Event	V4.0.0	Optional	listener: - POST hub: - POST - DELETE
TMF701	Process Flow	V4.0.0	Optional	processFlow: - POST - GET - GET /id - DELETE taskFlow: - GET - GET /id - PATCH

3.2. Dependent APIs

The following diagram illustrates API/Resource/Operation:



API ID	API Name	Mandatory / Optional	Operations	Rationales
TMF637	Product Inventory Management API	Mandatory	product: - GET - GET /id	Check required to verify any product inventory related to the party.
TMF620	Product Catalog Management API	Mandatory	catalog: - GET - GET /id category: - GET - GET /id productOffering: - GET - GET /id productOfferingPrice: - GET - GET /id productSpecification: - GET - GET /id	Product configuration must rely on product catalog information.
TMF622	Product Ordering Management API	Mandatory	productOrder: - GET - GET /id	Product configurator must produce a product order.
TMF632	Party Management API	Optional	individual: - GET - GET /id organization: - GET - GET /id	n/a
TMF662	Entity Catalog Management API	Optional	entityCatalog: - GET - GET /id	n/a
TMF666	Account Management API	Optional	billingAccount: - GET - GET /id	n/a
TMF669	Party Role Management API	Optional	partyRole: - GET - GET /id	n/a
TMF672	User Roles Permissions	Optional	permission: - GET - GET /id userRole: - GET - GET /id	n/a
TMF673	Geographic Address	Optional	geographicAddress: - GET - GET /id geographicSubAddress:	n/a

API ID	API Name	Mandatory / Optional	Operations	Rationales
	Management API		- GET - GET /id	
TMF674	Geographic Site Management API	Optional	geographicSite: - GET - GET /id	n/a
TMF688	Event	Optional	event: - GET - GET /id	n/a
TMF701	Process Flow	Optional	processFlow: - POST - GET - GET /id - DELETE taskFlow: - PATCH - GET - GET /id	n/a
TMF921	Intent Management API	Optional	intent: - GET - GET /id	n/a

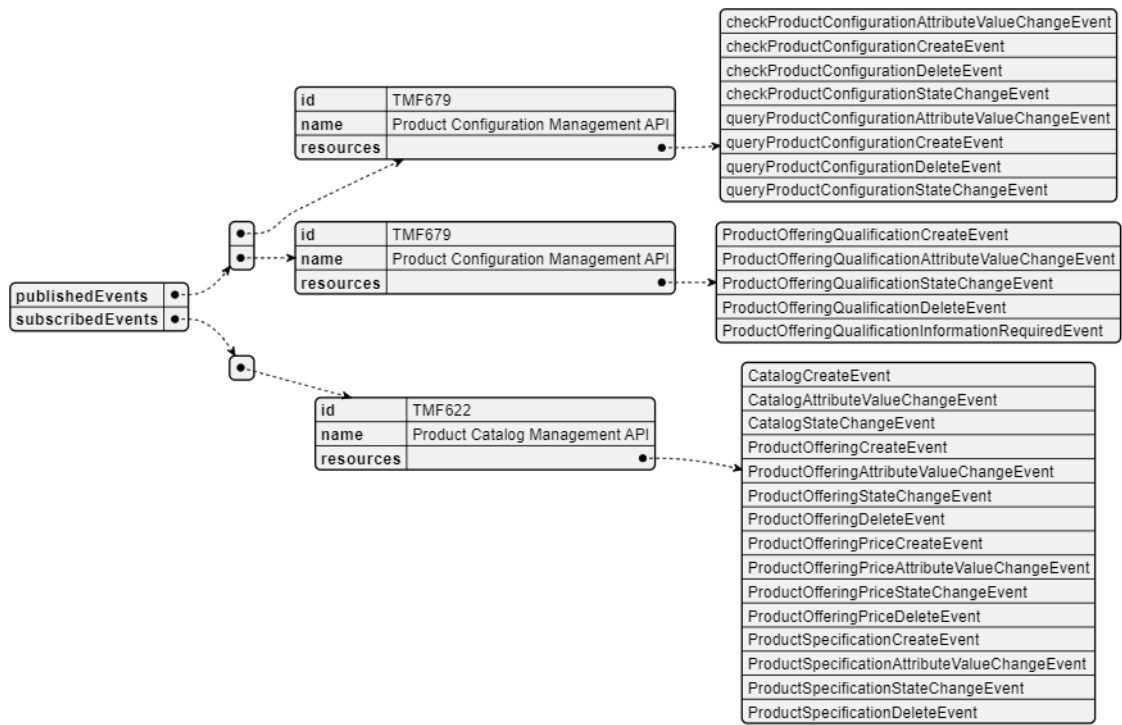
NOTE: Geographic Location Management API (TMF675) is available in Beta version. As soon as the interface will be published it will be added to the table and to the overview.

3.3. Events

The diagram illustrates the Events which the component publishes and the Events that the component subscribes to and then receives. Both lists are derived from the APIs listed in the preceding sections.

The type of event could be:

- Create : a new resource has been created (following a POST).
- Delete: an existing resource has been deleted.
- AttributeValueChange: an attribute from the resource has changed - event structure allows to pinpoint the attribute.
- InformationRequired: an attribute should be valued for the resource preventing to follow nominal lifecycle - event structure allows to pinpoint the attribute.
- StateChange: resource state has changed.



4. Machine Readable Component Specification

Refer to the ODA Component table for the machine-readable component specification file for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
SID	24.0
eTOM	24.0
Functional Framework	24.0

5.2. Further resources

1. IG1228: please refer to IG1228 for defined use cases with ODA components interactions.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
1.0.0	29 Mar 2022	Gaetano Biancardi , Anastasios Sarantis , Sylvie Demarest Matteo Destino Emmanuel A. Otchere	Final edits prior to publication
1.0.1	27 Jul 2023	Ian Turkington	No content changed, simply a layout change to match template 3. Separated the YAML files to a managed repository.
1.0.1	15 Aug 2023	Amaia White	Final edits prior to publication
2.0.0	10 Nov 2023	Sylvie Demarest	Add of TMF760 Product Configuration as exposed API. Check of dependent APIs. Improvement of the component description, and update of the mapping to Frameworks 23.0
2.1.0	24 Jun 2024	Gaetano Biancardi	- Updated to latest component template - Updated to odf V23.5 Exposed API, removed: - TMF688 Event Mgmt - TMF672 User Roles and Permissions Dependant API, removed - TMF688 Event Mgmt
2.1.1	18 Nov 2024	Gaetano Biancardi	API version, only major version to be specified
2.1.1	27 Dec 2024	Rosie Wilson	Final Edits Prior to Publication

6.1.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	29 Mar 2022	Goutham Babu	Initial release
Pre-production	15 Aug 2023	Amaia White	new release 1.0.1
Production	06-Oct-2023	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	20 Dec 2023	Amaia White	New release 2.0.0
Production	09-Feb-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-Production	12 Jul 2024	Amaia White	New release 2.1.0
Production	30-Aug-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-Production	27 Dec 2024	Rosie Wilson	New release 2.1.1
Production	07-Mar-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status

6.2. Acknowledgements

This document was prepared by the members of the TM Forum Component and Canvas project team:

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